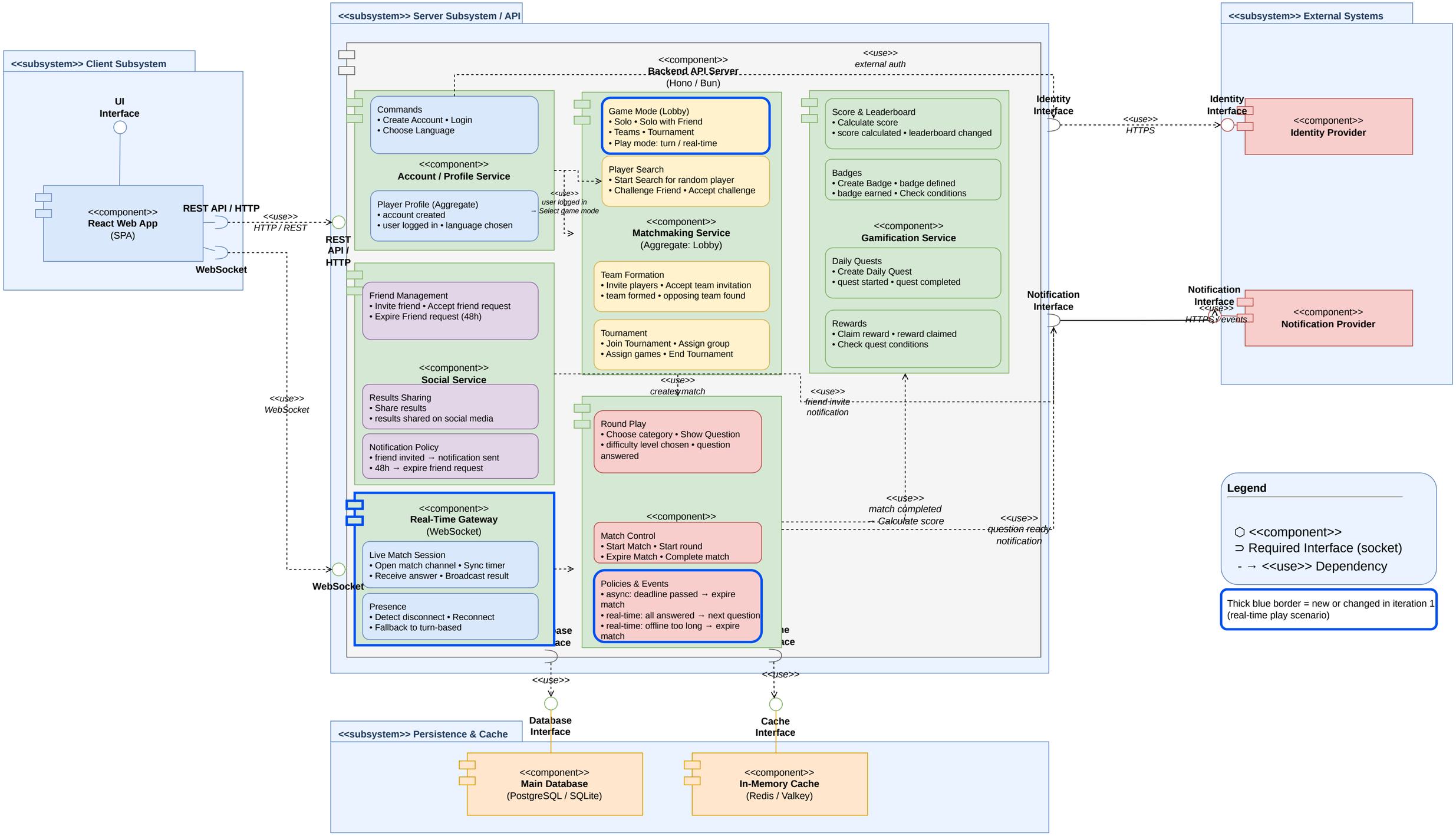

<<subsystem>> Client Subsystem
UI Interface
<<component>>
React Web App
(SPA)
REST API / HTTP
WebSocket
<<use>>
HTTP / REST
<<use>>
WebSocket
REST API / HTTP
<<subsystem>> Server Subsystem / API
<<component>>
Backend API Server
(Hono / Bun)
<<use>>
external auth
Commands
• Create Account • Login
• Choose Language
<<component>>
Account / Profile Service
Player Profile (Aggregate)
• account created
• user logged in • language chosen
<<use>>
user logged in
→ Select game mode
Game Mode (Lobby)
• Solo • Solo with Friend
• Teams • Tournament
• Play mode: turn / real-time
Player Search
• Start Search for random player
• Challenge Friend • Accept challenge
<<component>>
Matchmaking Service
(Aggregate: Lobby)
Team Formation
• Invite players • Accept team invitation
• team formed • opposing team found
Tournament
• Join Tournament • Assign group
• Assign games • End Tournament
Score & Leaderboard
• Calculate score
• score calculated • leaderboard changed
Badges
• Create Badge • badge defined
• badge earned • Check conditions
<<component>>
Gamification Service
Daily Quests
• Create Daily Quest
• quest started • quest completed
Rewards
• Claim reward • reward claimed
• Check quest conditions
Friend Management
• Invite friend • Accept friend request
• Expire Friend request (48h)
<<component>>
Social Service
Results Sharing
• Share results
• results shared on social media
Notification Policy
• friend invited → notification sent
• 48h → expire friend request
<<use>>
creates match
<<use>>
friend invite notification
Round Play
• Choose category • Show Question
• difficulty level chosen • question answered
<<component>>
Real-Time Gateway
(WebSocket)
Live Match Session
• Open match channel • Sync timer
• Receive answer • Broadcast result
Presence
• Detect disconnect • Reconnect
• Fallback to turn-based
<<component>>
Match Control
• Start Match • Start round
• Expire Match • Complete match
Policies & Events
• async: deadline passed → expire match
• real-time: all answered → next question
• real-time: offline too long → expire match
<<use>>
match completed
→ Calculate score
<<use>>
question ready notification
Identity Interface
<<use>>
HTTPS
<<subsystem>> External Systems
Identity Interface
<<component>>
Identity Provider
Notification Interface
<<use>>
HTTPS / events
Notification Interface
<<component>>
Notification Provider
WebSocket
Database Interface
Cache Interface
<<use>>
<<use>>
<<subsystem>> Persistence & Cache
<<component>>
Main Database
(PostgreSQL / SQLite)
<<component>>
In-Memory Cache
(Redis / Valkey)
Legend
<<component>>
Required Interface (socket)
<<use>> Dependency
Thick blue border = new or changed in iteration 1 (real-time play scenario)